

Chapter 7

The Yoneda Lemma

7.1 Hom-sets

Definition 7.1.1 (Hom-sets). For objects A and B in a category $\mathcal{C}$, a *hom-set* $\mathcal{C}(A, B)$ is a set of morphisms with domain A and codomain B .

Remark 7.1.2. Some authors write $\mathcal{C}(A, B)$ as $\text{hom}_{\mathcal{C}}(A, B)$ or $\text{Hom}_{\mathcal{C}}(A, B)$.

Definition 7.1.3 (Small Categories). A category $\mathcal{C}$ is *small* if its morphisms can be indexed by a set. $\mathcal{C}$ is *locally small* if $\mathcal{C}(A, B)$ can be indexed by a set for any pair of objects A and B .

Remark 7.1.4. Most categories that we use are locally small. However, functor categories are not always so.

Theorem 7.1.5. If $\mathcal{C}$ is small and $\mathcal{D}$ is locally small, then $\mathcal{D}^{\mathcal{C}}$ is locally small.

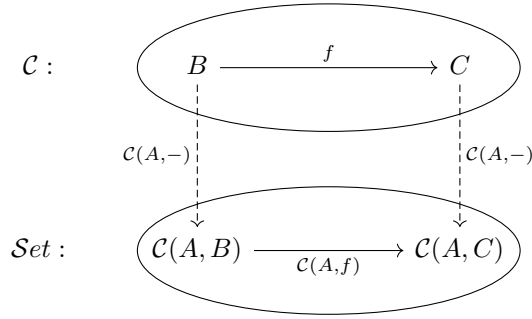
Proof. For a natural transformation $\alpha : \mathbf{F} \rightrightarrows \mathbf{G}$ between two functors $\mathbf{F}, \mathbf{G} : \mathcal{C} \longrightarrow \mathcal{D}$, we have to specify a morphism $\alpha_C : \mathbf{F}C \longrightarrow \mathbf{G}C$ in $\mathcal{D}$ for each object

C of $\mathcal{C}$. Thus there is a candidate for such a natural transformation α for every function

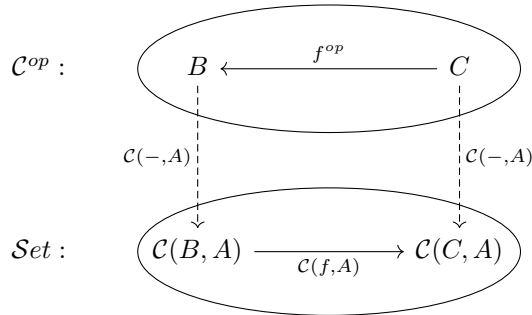
$$ob\mathcal{C} \longrightarrow \bigcup \{\mathcal{D}(\mathbf{FC}, \mathbf{GC}) \mid C \in ob\mathcal{C}\}$$

and there is certainly a set of such functions.¹ $\square$

Remark 7.1.6. If $\mathcal{C}$ is locally small, a hom-set $\mathcal{C}(A, B)$ is an object in $\mathbf{Set}$. Then we can define a covariant functor $\mathcal{C}(A, -) : \mathcal{C} \longrightarrow \mathbf{Set}$, which assigns an object B in $\mathcal{C}$ to $\mathcal{C}(A, B)$ in $\mathbf{Set}$ and a morphism $f : B \longrightarrow C$ in $\mathcal{C}$ to $\mathcal{C}(A, f) : \mathcal{C}(A, B) \longrightarrow \mathcal{C}(A, C)$ that maps a morphism $g : A \longrightarrow B$ in $\mathcal{C}(A, B)$ to $f \circ g$ in $\mathcal{C}(A, C)$.



Remark 7.1.7. Dually, we can define a contravariant functor $\mathcal{C}(-, A) : \mathcal{C}^{op} \longrightarrow \mathbf{Set}$, which assigns an object B in $\mathcal{C}^{op}$ to $\mathcal{C}(B, A)$ in $\mathbf{Set}$ and a morphism $f^{op} : C \longrightarrow B$ in $\mathcal{C}^{op}$ to $\mathcal{C}(f, A) : \mathcal{C}(B, A) \longrightarrow \mathcal{C}(C, A)$ that maps a morphism $g^{op} : B \longrightarrow A$ in $\mathcal{C}(B, A)$ to $f^{op} \circ g^{op}$ in $\mathcal{C}(C, A)$.



¹Crole (1993, p.61)

Remark 7.1.8. Moreover, we can define a functor $\mathcal{C}(-, -) : \mathcal{C}^{op} \times \mathcal{C} \longrightarrow \mathcal{S}et$, which assigns a pair of objects A and B to an object $\mathcal{C}(A, B)$ in $\mathcal{S}et$ and a pair of morphisms $f : A \longrightarrow A'$ and $g : B \longrightarrow B'$ to a morphism $\mathcal{C}(f, g)$ that maps a morphism h in $\mathcal{C}(A, B)$ to $h \circ f \circ g$.

Definition 7.1.9 (Yoneda functor). Since functor categories are cartesian closed, *Yoneda functor* is obtained though transposing the functor $\mathcal{C}(-, -) : \mathcal{C}^{op} \times \mathcal{C} \longrightarrow \mathcal{S}et$.

$$Y : \mathcal{C}^{op} \longrightarrow \mathcal{S}et^{\mathcal{C}}$$

There is also its dual:

$$Y' : \mathcal{C} \longrightarrow \mathcal{S}et^{\mathcal{C}^{op}}$$

7.2 Yoneda Lemma

Theorem 7.2.1 (Yoneda). Let $\mathcal{C}$ be a small category. If $\mathbf{K} : \mathcal{C} \longrightarrow \mathcal{S}et$ is a functor and A an object in $\mathcal{C}$, there is a isomorphism:

$$\mathcal{S}et^{\mathcal{C}}(\mathcal{C}(A, -), \mathbf{K}) \cong \mathbf{K}A$$

which sends each natural transformation $\alpha : \mathcal{C}(A, -) \xrightarrow{\sim} \mathbf{K}$ to $\alpha_A id_A$, and sends each element $a \in \mathbf{K}A$ to a natural transformation β whose component β_B sends $f \in \mathcal{C}(A, B)$ to $(\mathbf{K}f)a$.²

Remark 7.2.2. Recall that $\mathcal{S}et^{\mathcal{C}}$ is a category of functors from $\mathcal{C}$ to $\mathcal{S}et$. Thus $\mathcal{S}et^{\mathcal{C}}(\mathcal{C}(A, -), \mathbf{K})$ is a collection of natural transformations from $\mathcal{C}(A, -)$ to $\mathbf{K}$. By Theorem 7.1.5, this collection is a set.

Proof. Any element $a \in \mathbf{K}A$ is sent to a natural transformation $\beta \in \mathcal{S}et^{\mathcal{C}}(\mathcal{C}(A, -), \mathbf{K})$ and then sent back to $\beta_A id_A \in \mathbf{K}A$, but this equals to a as shown below:

$$\begin{aligned} \beta_A id_A &= (\mathbf{K}id_A)(a) \\ &= id_{\mathbf{K}A}(a) \\ &= a \end{aligned}$$

²MacLane (1997, p.61:(4))

Conversely, any natural transformation $\alpha \in \text{Set}^{\mathcal{C}}(\mathcal{C}(A, -), \mathbf{K})$ is sent to $\alpha_A id_A$ and then sent back to a natural transformation $\beta \in \text{Set}^{\mathcal{C}}(\mathcal{C}(A, -), \mathbf{K})$, but this equals to α since each component β_B sends $f : A \rightarrow B$ to:

$$\begin{aligned}
 (\mathbf{K}f)(\alpha_A id_A) &= (\mathbf{K}f \circ \alpha_A)(id_A) && \square \\
 &= (\alpha_B \circ \mathcal{C}(A, f))(id_A) && \text{(By naturality of } \alpha) \\
 &= \alpha_B(\mathcal{C}(A, f)(id_A)) \\
 &= \alpha_B(f \circ id_A) && \text{(By definition of } \mathcal{C}(A, f)) \\
 &= \alpha_B(f)
 \end{aligned}$$

There is another isomorphism for the dual Yoneda functor.

Corollary 7.2.3. Let $\mathcal{C}$ be a small category. If $\mathbf{K}' : \mathcal{C}^{op} \rightarrow \text{Set}$ is a functor and A an object in $\mathcal{C}^{op}$, there is a isomorphism:

$$\text{Set}^{\mathcal{C}^{op}}(\mathcal{C}(-, A), \mathbf{K}') \cong \mathbf{K}'A$$

which sends each natural transformation $\alpha : \mathcal{C}(-, A) \rightarrow \mathbf{K}'$ to $\alpha_A id_A$, and sends each element $a \in \mathbf{K}'A$ to a natural transformation β whose component β_B sends $f \in \mathcal{C}(B, A)$ to $(\mathbf{K}'f)a$.

Exercise 7.2.4. Prove the above corollary.

Theorem 7.2.5. Yoneda functor and its dual are full and faithful.

Proof. Apply $\mathcal{C}(B, -)$ to $\mathbf{K}$ of the Yoneda lemma. $\square$

7.3 Representable Functor

Definition 7.3.1 (Representation). Let $\mathcal{C}$ be a small category. A *representation* of a functor $\mathbf{K} : \mathcal{C} \rightarrow \text{Set}$ is a pair $\langle R, \psi \rangle$, with R an object of $\mathcal{C}$ and

$$\psi : \mathcal{C}(R, -) \cong \mathbf{K}$$

a natural transformation. The object R is called the *representing* object. The functor $\mathbf{K}$ is said to be *representable* when such a representation exists.³

³MacLane (1997, p.60:(3))

Theorem 7.3.2. Every universal arrow is a representable functor, and vice versa.

Proof. (to be written) □

Definition 7.3.3 (Representables).

7.4 End and coEnd

Theorem 7.4.1 (Yoneda Lemma as ends).

$$\int_{c' \in \mathcal{C}} \text{Set}(\mathcal{C}(c', c), \mathbf{K}c') \cong \mathbf{K}c$$

7.5 coYoneda Lemma

The following statement is called the coYoneda lemma by MacLane which is attributed to Kan.

Theorem 7.5.1 (coYoneda). Suppose that $\mathcal{C}$ is a category, $\mathbf{K} : \mathcal{C} \rightarrow \text{Set}$ is a functor:

$$\text{Set}^{\mathcal{C}}(\mathbf{K}, \mathcal{C}(A, -)) \cong \mathcal{C}^{(* \downarrow \mathbf{K})}(\Delta A, \pi)$$

Remark 7.5.2. $(* \downarrow \mathbf{K})$ is a subcategory of the comma category $(\text{Set} \downarrow \mathbf{K})$ whose object (f, X) consists of a morphism $f : * \rightarrow \mathbf{K}A$ for some $A \in \mathcal{C}$.

Recall that ΔA is a constant diagram defined in Definition 5.2.2.

$$\begin{array}{ccc} & * & \\ & \swarrow & \\ \mathbf{K}A & \longrightarrow & \mathbf{K}B \end{array}$$

³MacLane (1997, p.62:Ex)

There is an alternative statement for coYoneda:

Theorem 7.5.3 (coYoneda (alternative)). Every presheaf $\mathbf{K}$ is a colimit of representables, in that:

$$\mathbf{K} \cong \int^{c' \in \mathcal{C}} \mathcal{C}(c, c') \otimes \mathbf{K}c'$$

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